

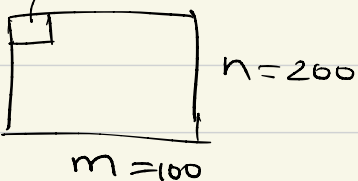
Generative Models

Problem Formulation :

Start with $(\Omega, \mathcal{F}, \mathbb{P})$, define a function $X(\text{R.v.}) : \Omega \rightarrow \mathbb{R}^d$

Once a R.v is defined, we'll have $(\mathbb{R}^d, \mathcal{B}, \mathbb{P}_X)$: work with this.

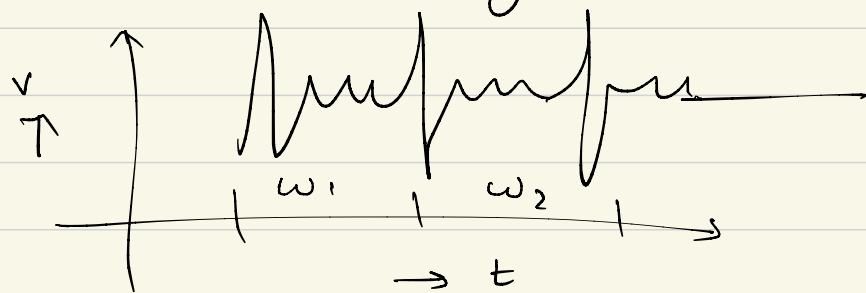
Consider a problem with images.

a)  $\therefore$ can be visualized as a vector in $\mathbb{R}^{mn}$.

point / vector in $\mathbb{R}^{20,000(d)}$

b) Tokens of natural language text
(word)
each word is a vector $\mathbb{R}^v$
 $v \rightarrow$ length of a dict.

c) speech signal



Each window is a vector $\mathbb{R}^w$
 $w \rightarrow$ length of the window.

Summary: Every datapoint is a
vector in an $\mathbb{R}^d$.

Suppose we have a R.V., (X)

$$X : \Omega \rightarrow \mathbb{R}^d$$

The datapoints (vectors in $\mathbb{R}^d$) are
the members of range space of
the function called the R.V.

Once we measure, we'll have

$$(\mathbb{R}^d, \mathcal{B}, P_X)$$

The probability measure ($\mathbb{P}$) completely specifies the underlying sample space.

Goal : Estimate the underlying prob. measure.

The above question translates to the following in practice:

practical Goal : Estimate the probability distribution function.

The inputs that are provided :

Given multiple datapoints:

$$D = \{x_1, x_2, \dots, x_N\}, \quad x_i \in \mathbb{R}^d$$

Eg : N-set of images, paragraphs, code-snippets.

x_i is a member of $\mathbb{R}^d$, which is the range space of a Rv X .

$\Rightarrow \exists$ a sample space Ω & the corresponding prob. measure P & a distribution function P_x .

$$D = \{x_i\}_{i=1}^n \sim P_x, \quad x_i \in \mathbb{R}^d.$$

Given D , estimate P_x .

In generative models, the problem is to not only estimate P_x , but also learn to sample from it.

Sampling: process of conducting the experiment that gave rise to Ω .

Problem Formulation for Gen. AI

Given a dataset D ,

$$D = \{x_1, x_2, \dots, x_N\} \sim \mathbb{P}_x \text{ (unknown)}$$

$$x_i \in \mathbb{R}^d,$$

objective: Estimate $\mathbb{P}_x$ & sample from it.

Framework for solving the Gen. AI problem.

i) Assume a parametric form on the unknown dist. function.

$\mathbb{P}_\theta$ where θ are the set of parameters

↳ "model" Eg: Neural Networks.

ii) Define / Measure a divergence metric b/w $\mathbb{P}_x$ & $\mathbb{P}_\theta$.

$D(P_x \parallel P_\theta)$: measures how far
are P_x & P_θ .

iii) Adjust the parameters θ , s.t
 $D(P_x \parallel P_\theta)$ is minimized.

$$\theta^* = \operatorname{argmin}_{\theta} \left[D(P_x \parallel P_\theta) \right]$$

iv) once P_x is estimate, use it to
sample.